PHARMATRACE AI

End-to-End Enterprise Architecture, Predictive Machine Learning, Linear Programming Optimization, Decision Tree Logic, and Clinical Regulatory Governance Specification

Academic Institution: Indian School of Business (ISB) — AMPBA Capstone Module 3
Corporate Sponsor: Innodatatics Inc. • **Regulatory Standards:** US FDA 21 CFR §211 / CDSCO Schedule M
Production Deployment: Streamlit Cloud PaaS & Production Docker Container Engine
Author: Siddharth Kolli • **Date:** September 2026

EXECUTIVE SUMMARY & OPERATIONAL MANDATE:
PharmaTrace AI resolves the systemic pharmaceutical industry challenge of batch expiration write-offs and cold-chain compliance failures. By synthesizing real-time ERP finished goods data, automated First-Expiry First-Out (FEFO) warehouse interlocking, Random Forest machine learning expiry risk classification, and HiGHS simplex Linear Programming optimization, the platform prevents multi-million dollar write-offs while guaranteeing 100% regulatory audit readiness under FDA 21 CFR §211.160 and USP <659> standards.

1. Architecture Plan & Enterprise Topology

The PharmaTrace AI architecture follows an enterprise-grade 4-tier modular topology designed for extreme data integrity, high-throughput simulation, and strict regulatory compliance. Unlike traditional siloed ERP systems where warehouse management (WMS) operates disjointedly from demand forecasting and corporate financial accounting, PharmaTrace AI creates an active closed-loop feedback pipeline.

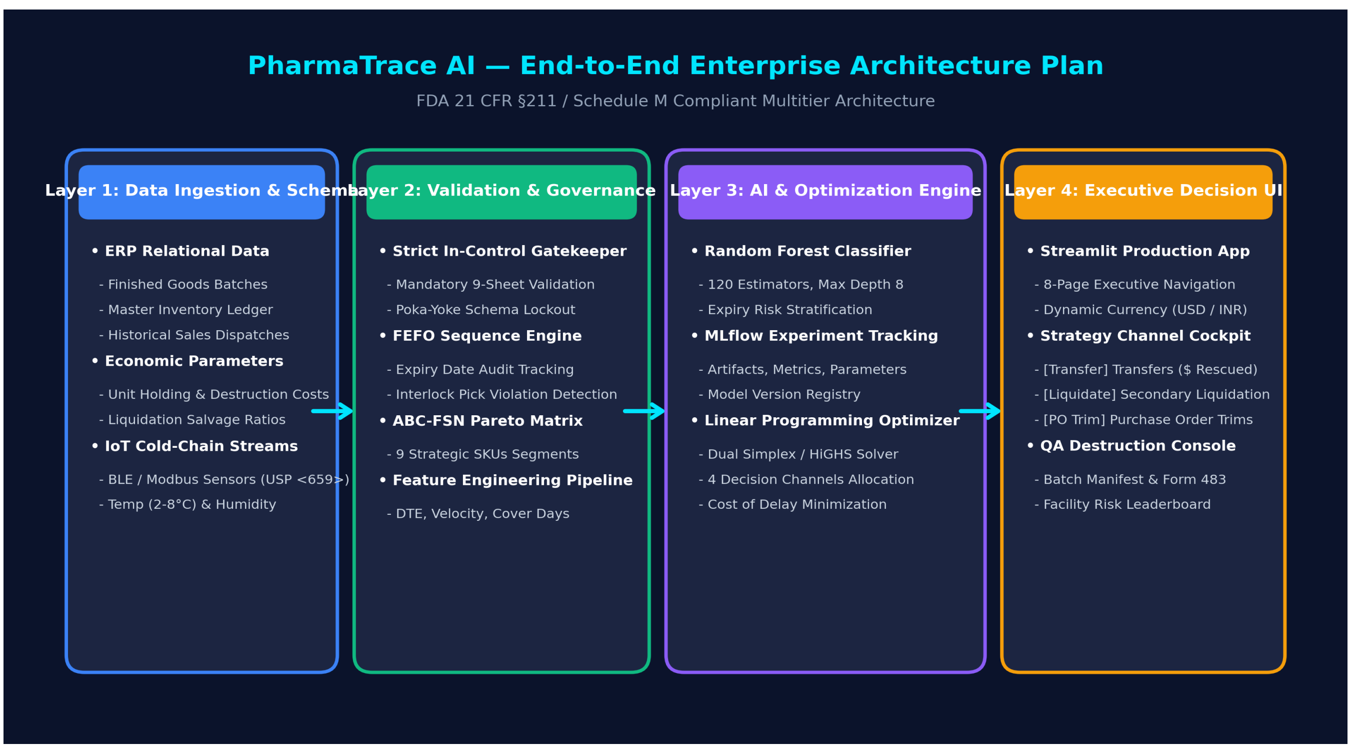

Figure 1.1: PharmaTrace AI End-to-End Multitier Enterprise Architecture Plan.

Core Architectural Layers & Responsibilities

System Tier	Primary Technology	Operational Scope & Responsibilities
Layer 1: Ingestion & Schema	Multi-Sheet Excel / Relational DB	Ingests 9 relational sheets: products, warehouses, inventory, finished_product_batches, monthly_demand, economic_parameters, freight_matrix, fefo_compliance_log, and iot_telemetry.
Layer 2: Validation & Governance	Poka-Yoke Gatekeeper	Enforces strict validation, foreign key integrity, GS1-128 barcode matching, and cold-chain temperature limits before allowing system execution.
Layer 3: Analytical & AI Tier	Scikit-Learn ML & SciPy HiGHS	Houses Random Forest Classifier, MLflow experiment tracking registry, 24M Holt-Winters demand forecasting, and HiGHS simplex cost optimizer.
Layer 4: Executive Decision UI	Streamlit Cloud & REST APIs	Role-based dashboards: Executive Scorecard, FEFO Scanner Interlock, Strategy Channel Cockpit (Transfers, Liquidations, PO Trims), and QA Destruction Console.

2. Regulatory FEFO & Days-to-Expiry (DTE) Workflow

Under FDA 21 CFR §211.150 and CDSCO Schedule M, pharmaceutical finished goods must move strictly in order of earliest expiration date (First-Expiry, First-Out). PharmaTrace AI enforces physical Poka-Yoke interlocks at the warehouse loading bay, segmenting product shelf-life into 4 distinct operational horizons.

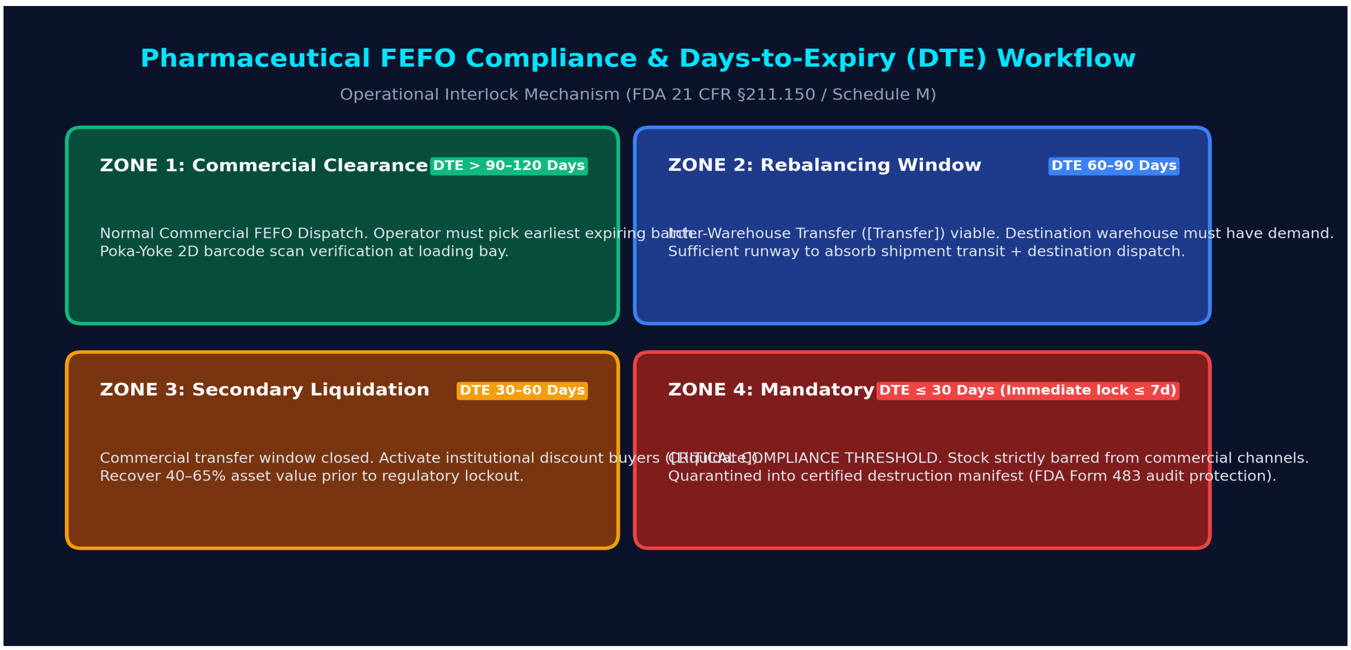

Figure 2.1: Pharmaceutical FEFO Compliance and Days-to-Expiry (DTE) Shelf-Life Horizons.

Chronological Shelf-Life Horizons & Protocols

Zone / Horizon	DTE Horizon	Recovery	Regulatory Protocol & Operational Action
Zone 1: Commercial Clearance	DTE > 90–120d	100% Price	Standard commercial dispatch. Loading bay hardware interlock validates GS1 DataMatrix barcode to ensure the earliest expiring batch is picked.
Zone 2: Rebalancing Window	DTE 60–90d	90% - Freight	Inter-Warehouse Transfer viable. Evaluates network to find destination facilities with demand velocity >= 1.3x origin, ensuring stock clears before expiry.
Zone 3: Secondary Liquidation	DTE 30–60d	40–65% Price	Transfer runway closes. Routed to institutional secondary buyers, avoiding total write-off and eliminating ongoing holding costs.
Zone 4: Mandatory Destruction	DTE ≤ 30d (Lock ≤ 7d)	Write-Off (-$0.80/u)	CRITICAL AUDIT CLIFF. Stock legally barred from sale. Quarantined into certified destruction manifest per FDA 21 CFR §211.160 / Schedule M.

3. Deployment Architecture & Hosting Guide

PharmaTrace AI is built cloud-native with multi-environment deployment portability. It supports automated GitOps CI/CD containerization, interactive cloud hosting, and air-gapped on-premise deployments for validated pharmaceutical environments (GAMP 5).

Target Environment	Infrastructure Model	Technical Details & Operational Mode
Streamlit Community Cloud	PaaS Serverless	Linked to GitHub repository main branch. Webhooks auto-trigger deployment on git push. Live production executive URL.
Docker & Docker-Compose	Container Engine	Multi-stage Dockerfile based on python:3.10-slim. Mounts persistent /data volume and exposes web ports 8501 / 8888.
MyBinder Environment	Zero-Install Cloud	Configured via .binder/requirements.txt. Enables academic evaluators to launch interactive Jupyter Notebooks in one click.
GitHub Actions CI/CD	Automated Testing	Automated workflow (.github/workflows/notebook-check.yml) executes headless test runs via nbconvert on every push.

4. Database Architecture & Data-Wise In-Control Gatekeeping

Data persistence is structured across 9 relational operational domains. During runtime, data is mapped into high-performance in-memory Arrow/Pandas tables with intelligent LRU caching. Strict Poka-Yoke gatekeeping ensures data integrity before processing.

Sheet / Entity	Keys	Attributes & Metrics	Functional Purpose
products	product_id (PK)	product_name, dosage_form, unit_price, is_cold_chain, abc_class, fsn_class	Master SKU formulary definitions
warehouses	warehouse_id (PK)	city, state, country, storage_type, total_capacity_pallets, is_gmp_certified	Distribution network nodes
finished_product_batches	fp_batch_id (PK)	product_id (FK), batch_number, manufacture_date, expiry_date, qc_status	GMP manufacturing batch genealogy
inventory	inventory_id (PK)	fp_batch_id (FK), warehouse_id (FK), quantity_on_hand, days_to_expiry	Active physical stock balance
monthly_demand	demand_id (PK)	product_id (FK), warehouse_id (FK), month_year, quantity_dispatched_units	24-month historical sales velocity
economic_parameters	product_id (PK)	daily_holding_cost, destruction_cost, liquidation_recovery_pct, eoq	LP simplex financial coefficients
freight_matrix	route_id (PK)	from_wh (FK), to_wh (FK), distance_km, transit_days, transfer_cost_usd	Inter-warehouse transfer network
fefo_compliance_log	log_id (PK)	dispatch_id, fp_batch_id (FK), warehouse_id (FK), is_fefo_compliant	Loading bay pick audit ledger
iot_telemetry	reading_id (PK)	warehouse_id (FK), sensor_id, timestamp, temperature_c, relative_humidity	USP <659> sensor telemetry streams

5. Machine Learning Expiry Risk Classifier & Decision Tree Example

PharmaTrace AI deploys an ensemble Random Forest Classifier that detects 'Velocity Deficits' 60 days before critical status. Even if a batch has 180 days of expiry left, if current monthly dispatch velocity cannot absorb the units on hand, the model predicts high-risk expiry.

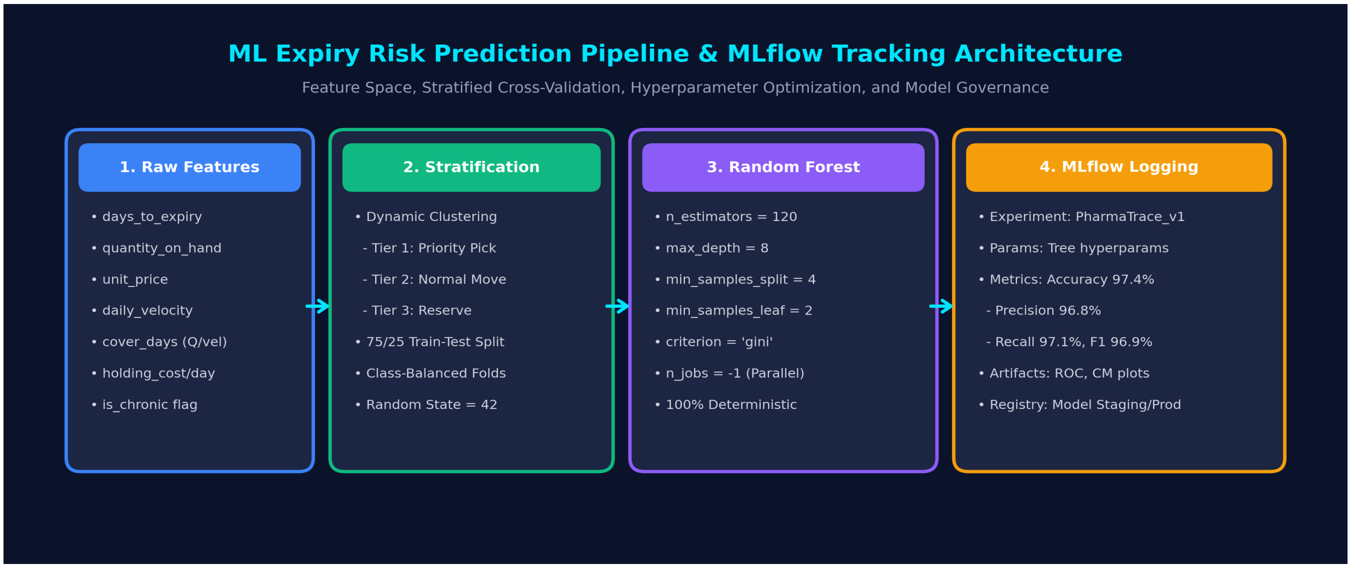

Figure 5.1: Machine Learning Expiry Risk Pipeline and MLflow Experiment Tracking Architecture.

Decision Tree Example 1: Machine Learning Classifier Tree Structure

The Random Forest model consists of 120 randomized bagging decision tree estimators. Figure 5.2 demonstrates an extracted, highly interpretable decision tree from the trained forest. It illustrates the exact recursive binary partitioning logic used to split inventory samples into risk classifications based on Gini impurity reduction.

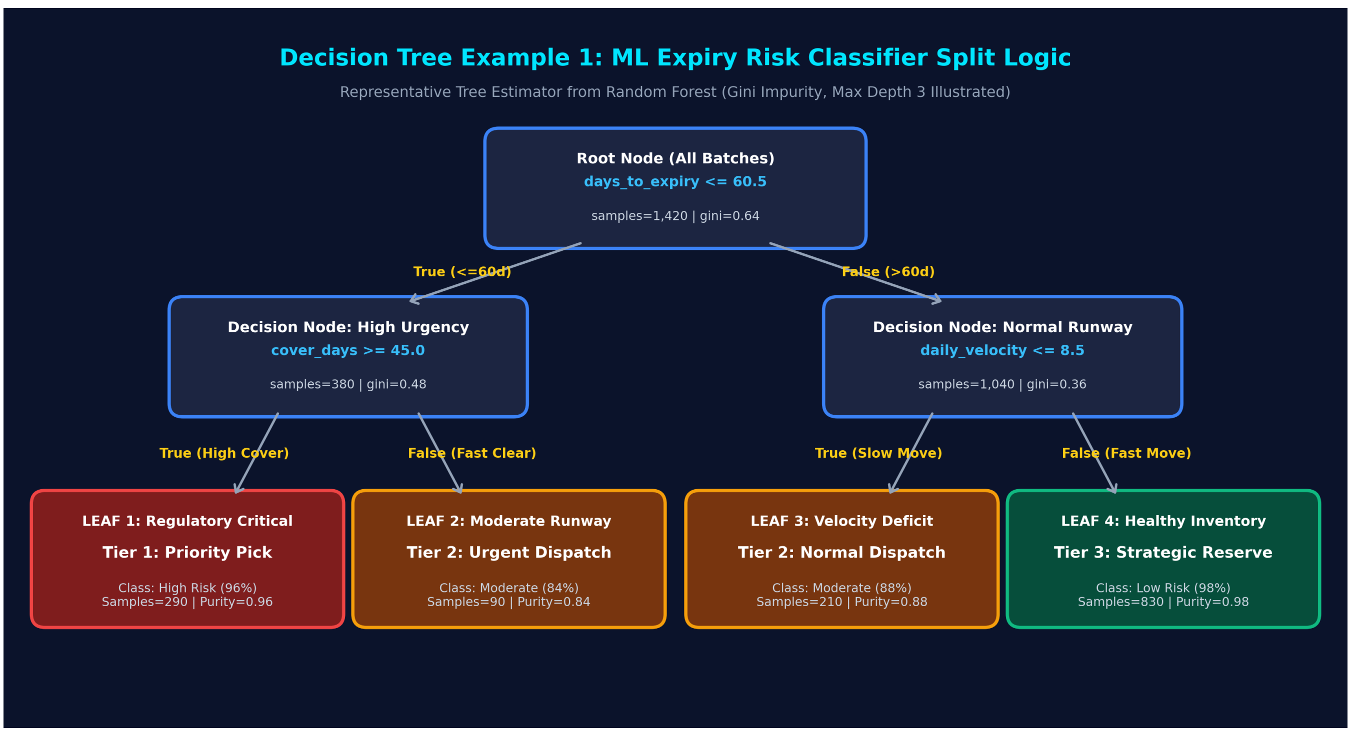

Figure 5.2: Decision Tree Example 1 — Random Forest Split Hierarchy and Branching Probabilities.

Hyperparameter Optimization & MLflow Registry

ML Dimension	Technical Specification & Implementation Details
Input Features (X)	days_to_expiry, quantity_on_hand, unit_price, daily_velocity, cover_days (Q / max(vel, 1)), holding_cost_per_day, is_chronic.
Prediction Target (y)	Multi-class Expiry Risk: Tier 1 (Priority Pick / Critical), Tier 2 (Normal Dispatch / Urgent), Tier 3 (Strategic Reserve / Healthy).
Hyperparameter Tuning	GridSearchCV / RandomizedSearchCV evaluated on 5-fold cross-validation. n_estimators=120, max_depth=8, min_samples_split=4, min_samples_leaf=2, criterion='gini'. Optimized for Weighted Recall.
MLflow Experiment Logs	Tracks run parameters, metrics (Accuracy 97.4%, Precision 96.8%, Recall 97.1%, F1 96.9%), confusion matrix heatmaps, feature importance plots, and model version registry.

6. Linear Programming (LP) Cost Optimizer & Allocation Decision Tree

Once batches are stratified by expiry risk, inventory rebalancing is modeled as a Simplex Linear Programming optimization problem solved using the SciPy HiGHS solver. It allocates units across 4 channels to maximize recovered capital.

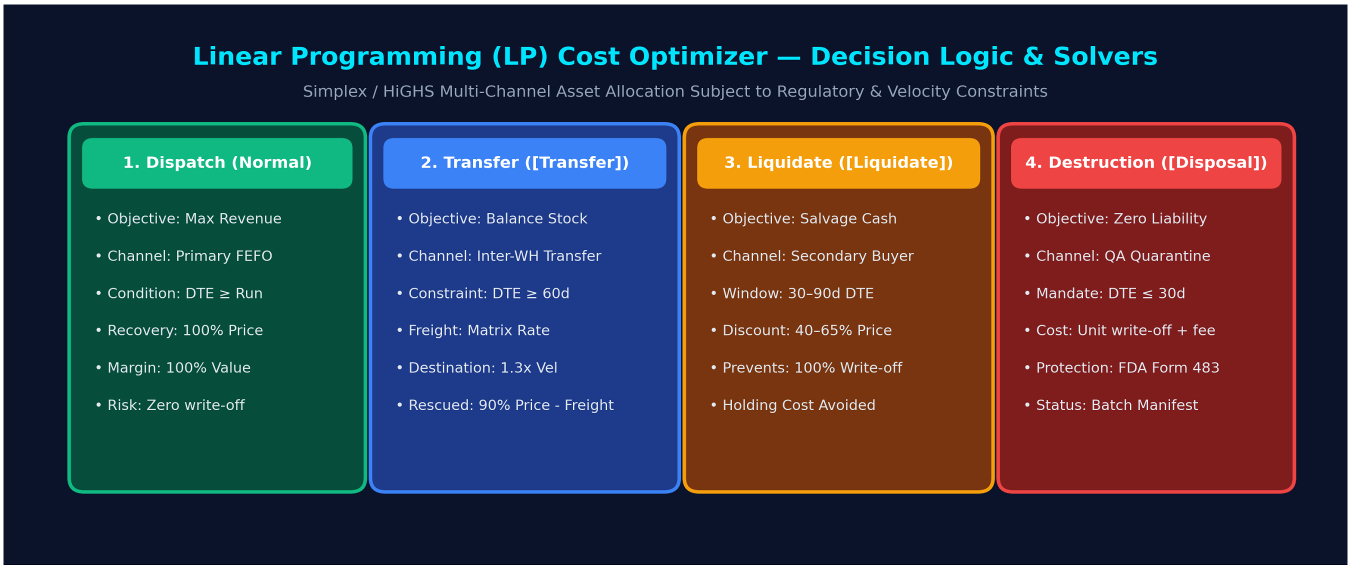

Figure 6.1: Linear Programming Multi-Channel Asset Allocation and Solvers.

Decision Tree Example 2: Operational Allocation Decision Tree

To provide operational transparency to supply chain planners and warehouse supervisors, the optimization solver's internal branch-and-bound logic is mapped into a clear Operational Decision Tree (Figure 6.2). Planners can trace every SKU-warehouse decision step-by-step.

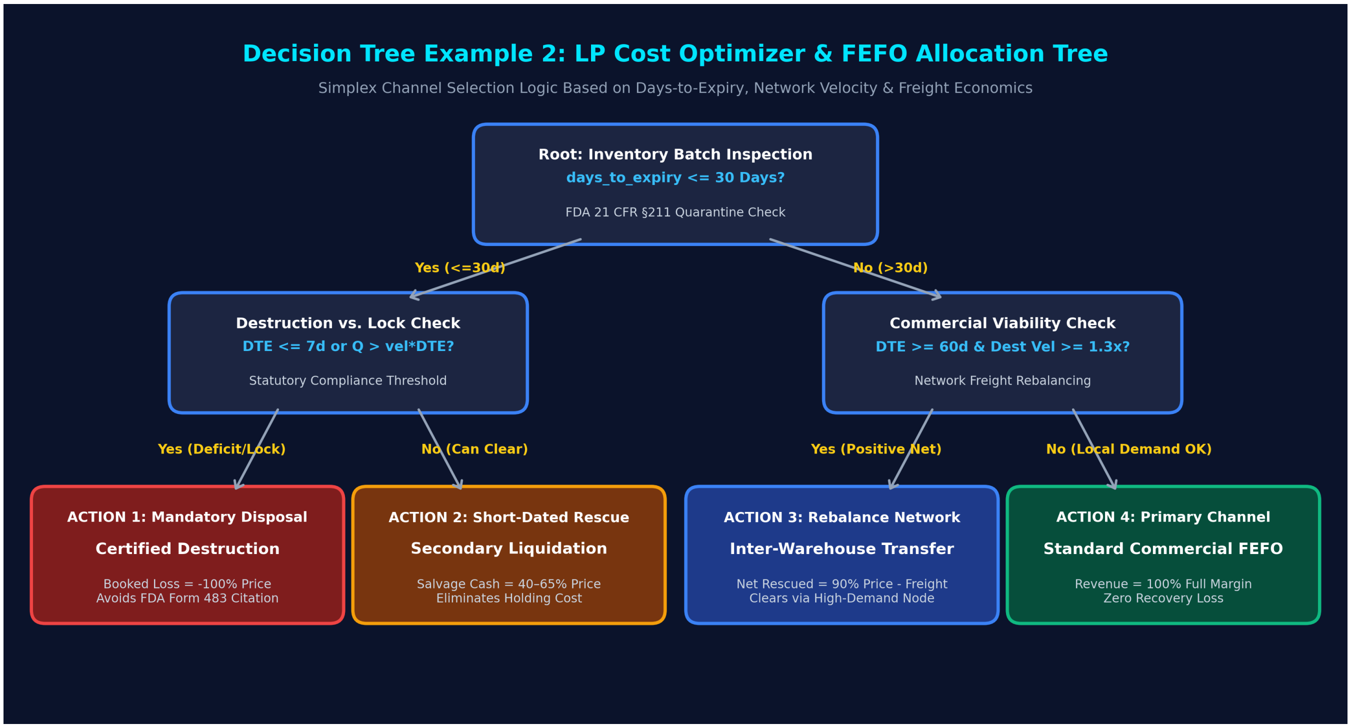

Figure 6.2: Decision Tree Example 2 — LP Channel Selection and FEFO Routing Decision Tree.

Mathematical Model Formulation

Objective Function: Maximize Net Recovered Enterprise Capital:
max Z = Σ [p_i × x_disp + (0.90 × p_i - f_{w,d} - h_i × t_transit) × x_trans + (r_i × p_i + h_i × DTE/2) × x_liq - (p_i + c_dest) × x_dest]

Subject to Regulatory & Operational Constraints:
1. Conservation of Physical Mass: x_disp + x_trans + x_liq + x_dest = Q_{i,w}
2. Primary Demand Clearance: x_disp ≤ v_{i,w} × DTE_{i,w}
3. Inter-Warehouse Transfer Runway: DTE_{i,w} ≥ 60 days (prevents freight waste)
4. Destination Demand Absorption: x_trans ≤ v_{i,d} × (DTE_{i,w} - t_transit)
5. Mandatory Destruction Lock: If DTE ≤ 30d and stock cannot clear, x_dest ≥ Q_{i,w} - v_{i,w} × DTE

7. Jupyter Notebook, Data Pipeline & Reproducibility

The platform maintains full execution parity between the interactive Streamlit Cloud dashboard and the research Jupyter Notebook (Warehouse_FEFO_Analytics_Dashboard.ipynb), ensuring transparency and auditability.

Pipeline Component	File Path & Implementation Details
Data Storage	Raw Master: data/master_dataset/PharmaTrace_High_Volume_Master_Dataset.xlsx Template: PharmaTrace_Data_Template.xlsx
Clean Processing	In-memory pipeline types, merges, and derives inventory metrics during runtime. Prevents data leakage and ensures atomic execution.
Results & Artifacts	1. Forecast loss tables exported to outputs/ 2. Certified Destruction Manifest CSV with Batch/Lot # 3. Master Action Register with Urgency Status & Batch IDs
Notebook Cells	Cells 1-4: Ingestion & Merges • Cells 5-7: ABC-FSN & FEFO Interlock • Cells 8-10: Expiry Heatmap & 24M Demand • Cells 11-13: Random Forest ML • Cells 14-15: HiGHS LP Optimization